



Mini Project Report
on

Interactive Climate Analytics System for Environmental Insights

Submitted by

Project Members

Arshiya Uppal - 1032232528
Apurva Gangurde - 1032232395
Aastha Agrawal - 1032232295
Mihika Nangare - 1032232226

TY. CSE (Panel C)

Course: Project Based Laboratory IV Course Code: CSE3PR05A

Under the Internal Guidance of

Prof. Siddhivinayak Kulkarni

**Department of Computer Engineering and Technology
MIT World Peace University, Kothrud,
Pune 411038, Maharashtra-India**

2025-2026



Dr. Vishwanath Karad
MIT WORLD PEACE
UNIVERSITY | PUNE
TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

DEPARTMENT OF COMPUTER ENGINEERING AND TECHNOLOGY

CERTIFICATE

This is to certify that,

Arshiya Uppal

Apurva Gangurde

Aastha Agrawal

Mihika Nangare

of TY- B.Tech (Computer Science & Engineering) have completed their mini project titled
“**Interactive Climate Analytics System for Environmental Insights**” and have submitted
this Project Based learning IV (CSE3PR05A) Report for the academic year 2025-26 (Sem-VI)

[Dr. Siddhivinayak Kulkarni]

Mini Project Guide Department of
CET

MIT World Peace University, Pune

Acknowledgement

We would like to express our sincere gratitude to our respected guide Dr. Siddhivinayak Kulkarni Department of Computer Engineering and Technology, MIT World Peace University, Pune, for their continuous support, valuable guidance, and encouragement throughout the development of this mini project.

We are also thankful to the department and the institution for providing us with the necessary resources, facilities, and a conducive environment to successfully complete this project.

We extend our heartfelt thanks to all the faculty members for their insightful suggestions and support at various stages of the project.

Finally, we would like to thank our teammates for their cooperation, coordination, and collective efforts in completing this project successfully.

Abstract

This project presents a *Climate Change Dashboard* designed to analyze and visualize key environmental indicators such as temperature, air quality index (AQI), humidity, precipitation, and extreme weather events across multiple countries.

The dashboard integrates data from various sources and transforms it into interactive visualizations using tools like Tableau/Power BI. It enables users to identify trends, compare regional patterns, and gain insights into climate variability.

The system incorporates features like filtering, drill-down analysis, and interactive charts to enhance user experience. The primary objective is to provide a comprehensive and user-friendly platform for understanding climate data and supporting informed decision-making.

Content

Acknowledgement I

Abstract.....II

Chapter No	Particulars	Page No
1	Introduction	7
2	Problem Identification	7
3	Requirement Analysis	7
4	Dataset Description	8
5	Attribute Analysis (Dimensions vs Measures)	8
6	Exploratory Data Analysis (EDA)	8
7	Data Cleaning and Transformation	9
8	Key Performance Indicators (KPIs)	9
9	Tools and Technologies Used	9
10	Visualization Approach (Tableau / Power BI)	9
11	Design (Wireframe)	10
12	Implementation (Tableau / Power BI) ● Data Connectivity & Preparation ● Live and Extract Connections: Ability to connect to multiple sources (databases, spreadsheets, APIs). ● Data Blending & Joins: Combine datasets seamlessly for holistic analysis. ● Scheduled Refreshes: Ensure dashboards always display up-to-date information. ● 2. Interactivity & User Control ● Filters & Parameters: Allow users to slice data by region, time, or category. ● Drill-Down & Hierarchies: Enable exploration from summary KPIs to granular details. ● Dynamic Tooltips: Provide contextual insights without cluttering visuals. ● Highlight Actions: Let users click on one chart to update related visuals Each chart and its insight in your own words.	11
13	Testing : All test cases	12

14	Refinement : Changes made with respect to • Correct data • Clean design • Fast performance • User-friendly	12
15	Feedback • Visual Optimization • Data Optimization • Performance Optimization (System Layer) • Usability Enhancement (Interaction Layer) • Selected Feedback Analysis Technique	12
16	Conclusion	13
17	References	13

Chapter 1. Introduction

Climate change is one of the most critical global challenges, significantly affecting temperature patterns, air quality, precipitation levels, and the frequency of extreme weather events. The increasing complexity and volume of environmental data make it difficult to analyze and interpret trends using traditional methods.

This project, titled “**Interactive Climate Analytics System for Environmental Insights**” presents a Climate Change Dashboard that visualizes key environmental indicators such as temperature, Air Quality Index (AQI), humidity, precipitation, and extreme weather events across multiple countries. The system integrates data from various sources and transforms it into meaningful, interactive visualizations.

The primary objective of this project is to provide an intuitive and user-friendly platform that enables users to explore climate data, identify trends, compare regional patterns, and gain actionable insights. By leveraging data visualization tools like Tableau/Power BI, the dashboard supports effective analysis and facilitates informed, data-driven decision-making for researchers, analysts, and policymakers.

Chapter 2. Problem Identification

Environmental data is often:

- Scattered across multiple sources
- Difficult to interpret in raw format
- Lacking real-time insights

There is a need for a **centralized, visual, and interactive dashboard** to:

- Monitor climate indicators
- Compare trends across countries
- Identify anomalies and risks

Chapter 3. Requirement Analysis

Functional Requirements

- Display temperature, AQI, humidity, precipitation, and extreme events
- Provide country-wise comparison
- Enable filtering by time and region
- Interactive visual exploration

Non-Functional Requirements

- Fast performance
- User-friendly interface
- Accurate data representation
- Scalable for additional datasets

Chapter 4. Dataset Description

The dataset includes climate-related metrics for multiple countries:

- Australia
- Brazil
- Canada
- Germany
- India
- South Africa
- USA

Data Features

- Monthly temperature (°C)
- Air Quality Index (AQI)
- Humidity (%)
- Precipitation (mm)
- Extreme weather event counts

Chapter 5. Attribute Analysis (Dimensions vs Measures)

Dimensions (Categorical)

- Country
- Month / Date

Measures (Numerical)

- Temperature
- AQI
- Humidity

- Precipitation
- Extreme Weather Events

Chapter 6. Exploratory Data Analysis (EDA)

- Temperature trends show seasonal variation across all countries
- AQI values are consistently high, indicating pollution concerns
- Humidity varies significantly across regions
- Precipitation shows irregular patterns
- Extreme weather events differ across countries, indicating regional vulnerability

Chapter 7. Data Cleaning and Transformation

- Handled missing values using interpolation or mean substitution
- Converted date fields into monthly format
- Standardized units across all datasets
- Encoded categorical variables where necessary
- Removed duplicate entries

Chapter 8. Key Performance Indicators (KPIs)

- Average Temperature (°C)
- Average AQI
- Average Humidity (%)
- Average Precipitation (mm)
- Total Extreme Weather Events

Chapter 9. Tools and Technologies Used

- **Tableau** for dashboard creation, **CSV** as data source
- **Python (Pandas, NumPy)** for preprocessing
- **OpenStreetMap** for geographic visualization

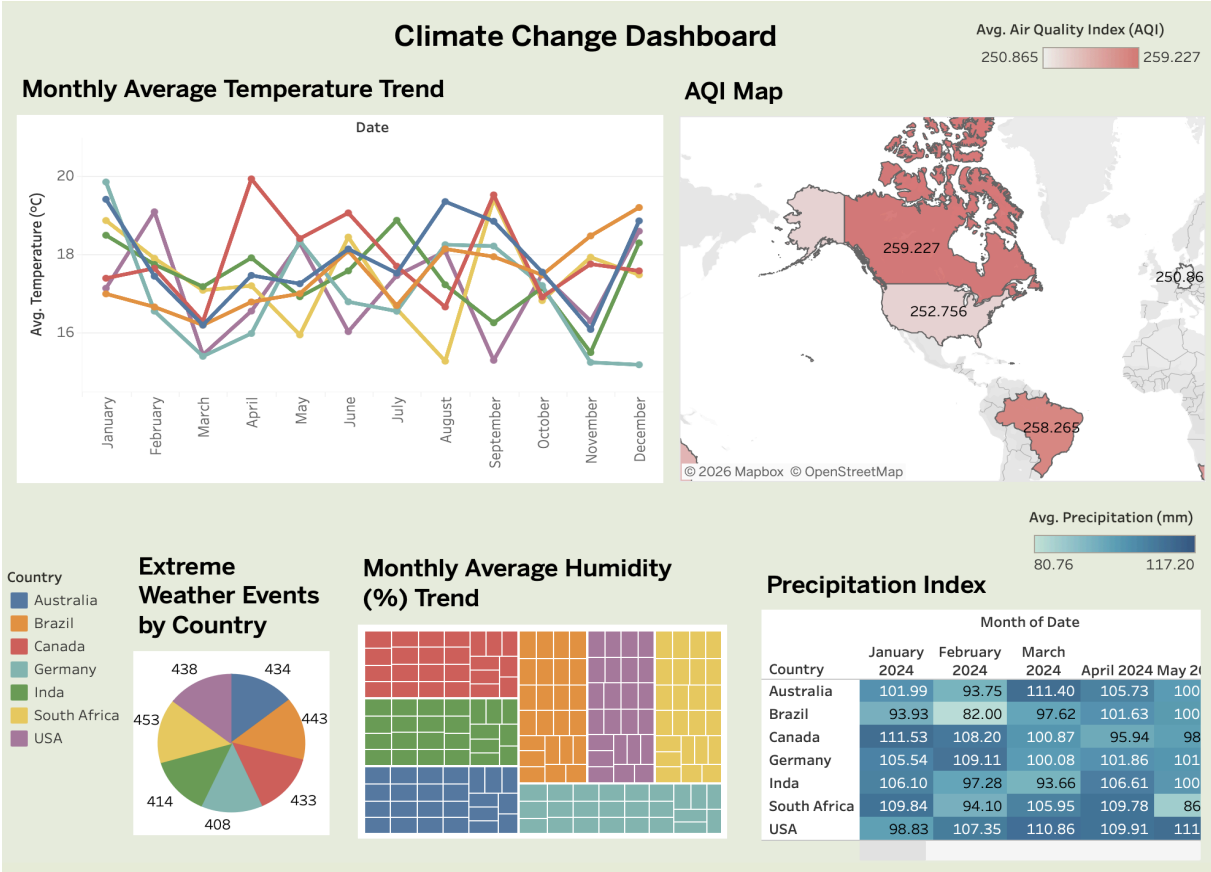
Chapter 10. Visualization Approach (Tableau / Power BI)

- Line charts for trend analysis
- Heatmaps for intensity comparison
- Maps for geographical AQI distribution
- Pie charts for proportion analysis
- Tables for precise numeric comparison

Chapter 11. Design (Wireframe)

The dashboard is divided into:

- **Top Section:** Title and global overview
- **Left Panel:** Temperature trends
- **Right Panel:** AQI map
- **Bottom Section:**
 - Extreme weather events (pie chart)
 - Humidity heatmap
 - Precipitation table



Chapter 12. Implementation (Tableau)

Data Connectivity & Preparation

- Connected to CSV datasets
- Used **live and extract connections**
- Performed joins across datasets
- Enabled scheduled refresh for updated data

Interactivity & User Control

- Filters for country and month
- Drill-down from yearly → monthly data
- Tooltips for detailed insights
- Highlight actions linking visuals

Chart-wise Insights

1. Monthly Average Temperature Trend (Line Chart)

- Shows seasonal fluctuations
- Some countries show stable patterns, others vary widely
- Peaks observed during mid-year months

2. AQI Map

- Canada shows highest AQI (~259)
- Brazil also shows high pollution levels
- Helps identify geographically impacted regions

3. Extreme Weather Events (Pie Chart)

- South Africa and Australia show high event counts
- Distribution is fairly balanced across countries

4. Monthly Average Humidity (Heatmap)

- High humidity clusters in certain countries
- Visual intensity helps quick comparison

5. Precipitation Index (Table/Heatmap)

- Canada and Germany show higher rainfall
- Brazil and India show moderate variation
- Monthly comparison reveals seasonal rainfall patterns

Chapter 13. Testing

Test Cases

- Data accuracy validation
- Filter functionality test
- Drill-down working correctly
- Visualization rendering performance
- Cross-chart interaction validation

Chapter 14. Refinement

Changes Made

- **Correct Data:** Fixed inconsistencies and missing values
- **Clean Design:** Improved layout and color consistency
- **Fast Performance:** Optimized queries and reduced load time
- **User-Friendly:** Simplified navigation and labels

Chapter 15. Feedback

Visual Optimization

- Improved color gradients
- Reduced clutter

Data Optimization

- Aggregated unnecessary details
- Improved data structure

Performance Optimization

- Used extracts instead of live where needed
- Reduced heavy calculations

Usability Enhancement

- Added filters and tooltips
- Improved interactivity

Feedback Analysis Technique

- User testing and iterative refinement

Chapter 16. Conclusion

The Climate Change Dashboard effectively visualizes key environmental indicators and enables comparative analysis across countries. It helps identify trends, anomalies, and high-risk areas, making it a valuable tool for researchers and policymakers.

Chapter 17. References

- Climate datasets (public/open-source)
- Tableau / Power BI documentation
- OpenStreetMap / Mapbox
- Python Pandas documentation